Blood Flow Simulation Algorithm

This algorithm simulates unsteady, incompressible, non-Newtonian blood flow through arteries using the Bird-Carreau model on a staggered grid with the SIMPLE algorithm implemented in CUDA.

Algorithm Steps

1. Initialization Phase
   - ✓ Define physical parameters for blood (density, viscosity parameters)
   - ✓ Set up computational domain (artery dimensions, grid resolution)
   - ✓ Allocate memory for velocity, pressure, viscosity, and intermediate variables
   - ✓ Initialize flow velocity profile at inlet
   - ✓ Set initial pressure gradient and viscosity distribution

2. Time Advancement Loop
   - ⊗ For each time step until reaching the desired physical simulation time:
   - ⊗ Store Previous Values - Save current velocity fields for residual calculation
3. SIMPLE Algorithm Iteration:

   Repeat until convergence or maximum iterations:

   Apply Boundary Conditions:

   - o Enforce parabolic velocity profile at inlet
   - o Set zero gradient at outlet for velocity
   - o Apply no-slip condition at walls
   - o Set pressure boundary conditions

   Momentum Prediction:

   - o Solve x-momentum equation using QUICK scheme for convection
   - o Solve y-momentum equation using QUICK scheme for convection
   - o Use central differencing for diffusion terms
   - o Calculate intermediate velocity fields (u* and v*)

Calculate Divergence

⇒ Compute velocity field divergence for pressure correction

Pressure Correction:

⇒ Solve pressure Poisson equation using Gauss-Seidel with SOR
⇒ Apply appropriate boundary conditions for pressure

Velocity Correction:

- Update velocities using pressure gradient
- Ensure mass conservation

Update Viscosity:

∑ Calculate shear rate from velocity gradients
∑ Update viscosity using Bird-Carreau model

Check Convergence:

➢ Calculate residual between iterations
➢ Exit loop if convergence criteria met

Adaptive Time Step Calculation:

➢ Calculate maximum velocity in domain
➢ Adjust time step based on CFL condition
➢ Ensure numerical stability

4. Output Results
   ❖ Periodically save flow field data to VTK files
   ❖ Record simulation statistics (residuals, velocities)
   ❖ Generate visualization data for ParaView
5. Finalization
   ✓ Write final output files
   ✓ Free allocated memory
   ✓ Report simulation performance metrics